Machine Learning
Course Project Guidelines
A practical application for machine learning

We live in a world flooded with data. Globalization, widespread internet and mobile applications have converted us into a continuous source of information for big corporations, and also has given us the opportunity to access knowledge in a way that had never been possible before. What can you use artificial intelligence for? the answer is: pretty much the same that you can use your intelligence for.
So, the question would be: what do YOU want to use AI for?
In this project you will find a practical application of machine learning, something that motivates you and is relevant to people out there. Then, you will use the tools learnt during our course and provide a solution that satisfies both your appetite for knowledge and the needs of the potential users.

There will be two deliverables, the first one will be a "project proposal" which will be delivered on the 6th session. The second will be a scientific paper, in which you will communicate effectively the results of your research.

Group work is permitted in groups of up to four people.

First deliverable: Project Proposal

You will make a presentation with the following sections:

1. Introduction and motivation:
    - What is the problem (application) you will tackle?
    - Why is it important?
2. Literature review about other solutions to the problem
3. Your solution/process description, including an schematic diagram
4. Dataset explanation and visualization

You will submit the slides of your presentation. You will have 15 minutes to present and 5 mins for questions.

Final deliverable: Results Publication

You will write a scientific paper in IEEE format (in English) that contains at least the following items:

1. Description of the phenomenon / process modeled and the problem to be addressed
2. Process of obtaining / generating the dataset.
3. Description of the machine learning problem, what kind of problem is it? What is the learning task to perform?
4. Data pre-processing used using at least three different methodologies
5. Training and evaluation of at least three models, including a complexity analysis

In addition to the article, you will provide all the files needed to run your experiment, and make a practical presentation (20 minutes) in which you will demonstrate how it works and what were the results.

All project ideas are welcome. However, datasets included in Scikit Learn are not permitted.

Grading for the final deliverable

- Each working group should define a role for each member according to the next table.

Role	Key Responsibilities
Data Engineer	Clean data, handle missing values, features
Model Developer	Train/test models, tune hyperparameters
Project Manager	Timeline, GitHub repo, team check-ins
Evaluation Lead	Metrics, error analysis
Reporting Lead	Report writing, visuals, final slides

- Each group should use project manager tool like Trello (https://www.kaggle.com/discussions/getting-started/133275 - https://trello.com/b/cEdSEMvx).
- Each member should rate each teammate's contribution.

Each member will be graded according to the next criteria for the first and the final deliverable:

- **First/Final deliverable: 70%**
- **Report from the project manager tool: 20%**
- **Peer Review: 10%**